

- 6 (a) (i) Photon energy = eV_s + work function

$$\frac{hc}{\lambda} = 1.60 \times 10^{-19} \times 2.3 + \phi \quad \text{--- (1)}$$

$$\frac{hc}{620 \times 10^{-9}} = 1.60 \times 10^{-19} \times 1.1 + \phi \quad \text{--- (2)}$$

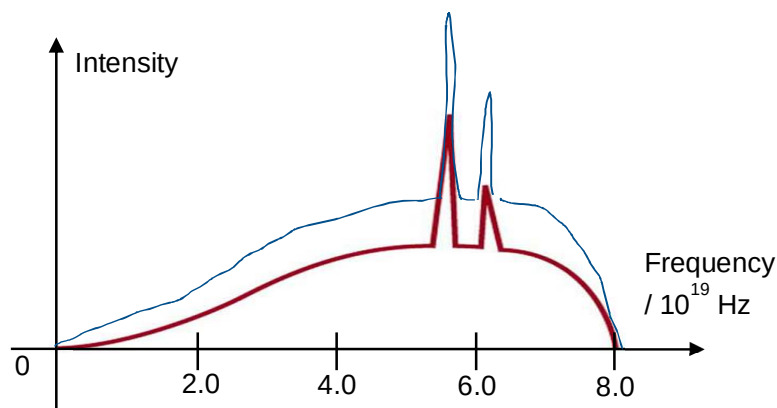
$$(1) - (2): hc \left(\frac{1}{\lambda} - \frac{1}{620 \times 10^{-9}} \right) = 1.60 \times 10^{-19} \times 1.2$$

$$\lambda = 3.87 \times 10^{-7} \text{ m}$$

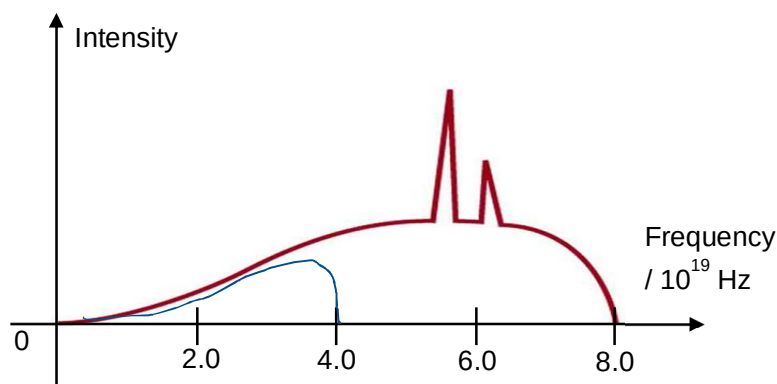
- (ii) For the same incident power, the rate of photon incident on the surface is proportional to the wavelength of the radiation. Since the unknown radiation has a shorter wavelength, the rate of photons incident on the silver surface is less and hence less photoelectrons are ejected, giving rise to a lower saturation current.
- (b) (i) When an accelerated electron loses all its KE to the target atom, the KE lost is converted into an photon of maximum energy, hence maximum frequency.

$$\begin{aligned} \text{(ii)} \quad E &= hf = (8.0 \times 10^{19})(6.63 \times 10^{-34}) \\ &= 5.304 \times 10^{-14} \text{ J} \\ &= 0.332 \text{ MeV} \end{aligned}$$

- (iii) sketch a graph with similar shape above original graph with same cut-off frequency.



- (iv) sketch a graph with similar shape below original graph without any characteristics lines with cut-off frequency at $4.0 \times 10^{19} \text{ Hz}$.



(c) (i)
$$\frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{436 \times 10^{-9}} = 4.56 \times 10^{-19} = 2.85 \text{ eV}$$

Since $E_4 - E_2 = 2.85 \text{ eV}$, the transition is from E_4 to E_2

- (ii) The bombarding electron must have a minimum kinetic energy of $(13.6 \text{ eV} - 0.54 \text{ eV})$ 13.06 eV in order for the hydrogen atom to be excited from ground state E_1 to E_4 . Hence a transition from E_4 to E_2 can take place. So the minimum value is 13.06 V.

7 (a) (i) Consider each mass separately